

Tirth Bhanderi

Rajkot, Gujarat, India

+91 9737006181 | bhanderitirth94@gmail.com | linkedin.com/in/tirth-bhanderi-345763289

Summary

AI and ML Engineering student and proficient Android and Flutter Developer with experience in Java, Python, and Spring Boot development.

Experience

Backend Engineering Intern | DEZAI, UAE (Remote)

June 2026 – Present

- Contributing to backend engineering for DEZAI's core product as part of a UAE-based startup team, building and maintaining server-side services and APIs using Node.js and NestJS.
- Collaborating with the founding team to design scalable backend architecture and ship product features in a fast-paced startup environment.
- Technologies:** Node.js, NestJS, REST APIs.

AI Full-Stack Engineer (Internal Tools) | Hytech Education

May 2026 – June 2026

- Directly appointed by the Founder to design and build an internal GenAI-powered compliance tool for Hytech Education, delivered within the stipulated one-month period.
- Built the solution end-to-end across the AI and full-stack layers; the tool is currently in active internal use.
- Technologies:** Python, GenAI/LLM integration, Full-Stack Development.

Android Developer Intern | 9Brainz, Rajkot, Gujarat

December 2024 – April 2025

- Developed and enhanced **Openly**, a full-featured social media app with real-time one-to-one and group video/audio calling using WebRTC, chat with image sharing, and media post interactions.
- Integrated Firebase Authentication, Cloud Firestore, and Cloud Storage; built profile editing, follow/unfollow, user discovery, bookmarking, and push notifications.
- Technologies:** Flutter (Dart), Firebase Authentication, Cloud Firestore, Firebase Storage, WebRTC.

Education

Bachelor of Technology in Artificial Intelligence and Machine Learning

Expected 2028

CHARUSAT

GPA: 8.23 (till 2nd Year)

Diploma in Computer Engineering

September 2023 – April 2025

RK University

GPA: 9.63

Technical Skills

Programming Languages: Java, Python, Dart, JavaScript, SQL

Frontend: React, Next.js, Tailwind CSS

Mobile Development: Flutter, Android SDK, WebRTC, REST API integration

Backend & Frameworks: Spring Core, Spring Boot, FastAPI, Flask, Node.js, NestJS, Microservices Architecture

Databases: MySQL, PostgreSQL, MongoDB, Firebase (Firestore, Realtime DB, Authentication, Storage), SQLite

AI / ML: TensorFlow Lite, CNN (MobileNet), Retrieval-Augmented Generation (RAG), LangChain

Tools & DevOps: Git, GitHub, Docker, Alembic

Projects

Krush-Netra

- Offline AI-powered Android app classifying 12 Indian cattle breeds using a MobileNet-based CNN model – optimized for low-spec devices in rural areas without internet access.
- Trained a CNN model to a lightweight TensorFlow Lite format, minimizing app size and inference latency for entry-level Android hardware.
- Built a simple camera-capture UI that returns the predicted breed with a confidence score entirely on-device, with no internet connectivity required.
- Technologies:** Android (Java), TensorFlow Lite, CNN, MobileNet.

Metis – AI-Powered Recruitment Platform

- Automates candidate evaluation via resume parsing (30%) and live contextual AI interviews (70%) with WebSocket-based voice/text support and leaderboard analytics.
- Role-based system for candidates and HR with job posting, transcript review, and scoring dashboard.
- Technologies:** Next.js, TypeScript, Tailwind CSS, Flask, MongoDB, Socket.IO, JWT, Groq API, LangGraph, Docker.

Charaka Vaidya – RAG-Powered Ayurvedic AI Assistant

- Clinical-grade AI assistant grounded in the Charaka Samhita using RAG with Sthana-weighted semantic reranking and mandatory Vata/Pitta/Kapha diagnostic structure.
- Multilingual (English, Hindi, Gujarati); includes Safety Filter, Intent Classifier, herb glossary, Dosha quiz, and PDF report generation.
- Technologies:** Python, React, LangChain, ChromaDB, HuggingFace Embeddings, Groq API (LLaMA 3 70B), RAG.

ML Model Inference Serving Platform

- Built a backend platform for deploying trained ML models (ONNX/TFLite) as production-style REST inference endpoints, with automatic input/output schema introspection and a model registry supporting versioning and staged rollout (staging → production).
- Designed a generic inference pipeline via a Redis-backed job queue decoupling the API gateway from a Python model worker, with task-specific postprocessing for classification (label/confidence ranking) and object detection (bounding box decoding, per-class Non-Max Suppression).
- Built a React frontend with an interactive canvas-based bounding box visualizer and a confidence-threshold slider for live result filtering, plus a stats dashboard tracking per-model latency and error rates.
- Technologies:** Spring Boot, PostgreSQL, Redis, Python (FastAPI), ONNX Runtime, TensorFlow Lite, React, Docker.